CSE221 Assignment 03 Fall 2026

A. Count the Inversion

1 second, 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Here is a Pseudocode of the Merge Sort Algorithm.

```
def merge(a, b):
    # write your code here
    # a and b are two sorted list
    # merge function will return a sorted list after merging
a and b

def mergeSort(arr):
    if len(arr) <= 1:
        return arr
    else:
        mid = len(arr)//2
        a1 = mergeSort(............)  # write the parameter
        a2 = mergeSort(............)  # write the parameter
        return merge(a1, a2)          # complete the merge
function above
```

Now, you are given an array ***A*** of size ***N*** of ***N*** distinct integers. It is guaranteed that the array A contains a permutation of integers from 1 to N (i.e., every integer from 1 to N appears exactly once).

1. Count the number of inversions in the given array.
2. Sort the array in non-decreasing order.

An inversion is a pair (i, j) where $i < j$ and $A[i] > A[j]$.

Input

The first line contains an integer ***N*** $(1 \leq N \leq 10^5)$ — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq N)$ separated by spaces.

Output

In the first line, print the total number of inversions in the given array. In the next line, print the array in non-decreasing order.

input

```
5
1 2 5 4 3
```

output

```
3
1 2 3 4 5
```

input

```
5
1 2 3 4 5
```

output

```
0
1 2 3 4 5
```

input

```
5
5 4 3 2 1
```

output

```
10
1 2 3 4 5
```

input

```
7
6 4 2 5 7 3 1
```

output

```
14
1 2 3 4 5 6 7
```

In the first example (1-based indexing), the inversions are the pairs of indices $(3,\ 4), (3,\ 5)$ and $(4,\ 5)$.

In the second example, there are no inversions.

In the third example, every pair of i, j where $i < j$, we have $A[i] > A[j]$. Hence, All 10 such pairs are inversions.

B. Count the Inversion Revisited

1 second, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given an array A of N integers. Find the number of pairs of indexes (i, j) such that $i < j$ and $A[i] > A[j]^2$.

Input

The first line contains an integer N $(1 \leq N \leq 10^5)$ — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \ldots a_n$ $(-10^6 \leq a_i \leq 10^6)$ separated by spaces.

Output

Output a single integer - number of such pairs.

input

```
6
10 2 5 1 -2 25
```

output

```
6
```

input

```
5
5 4 3 -2 -1
```

output

```
4
```

C. Fast Power Drift

1 second, 256 megabytes

Forbidden Keywords for the Quiz: pow, open, file, creat(, fstream, thread, process, system(, exec(

You are given two integers ***a*** and ***b***. Calculate $a^b \ mod \ 107$.

Input
The input file contains two integers ***a*** $(1 \le a \le 10^4)$ and ***b*** $(1 \le b \le 10^{12})$.

Output
Print one integer — the result of $a^b \ mod \ 107$.

input
100 3
output
85

input
100 5
output
99

input
10000 1000000000000
output
27

input
2 1 1 1 0 5 2 0 0 2 3
output
8 5 5 3 8 0 0 8

input
3 1 1 1 1 10 1 2 3 4 12 1 2 1 2 5
output
512 512 512 512 138067399 201223170 301834755 439902154 81 162 81 162

D. Fast Matrix Drift

1 second, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a 2×2 integer matrix A and an integer exponent X $(1 \le X \le 10^9)$.

Your task is to compute the matrix A^X, where all intermediate operations (additions and multiplications during matrix multiplication) are performed modulo $10^9 + 7$.

Formally, for two matrices:

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

Their product $D = B \times C$ is defined as:
$D_{ij} = (b_{i1} \cdot c_{1j} + b_{i2} \cdot c_{2j}) \bmod (10^9 + 7)$.

Input
The first line contains a single integer T $(1 \le T \le 10^5)$ — the number of test cases.

The first line of each test case contains four integers a_{11}, a_{12}, a_{21}, and a_{22} where $0 \le a_{ij} \le 10^9$ — the elements of the matrix A. The second line contains an integer X where $1 \le X \le 10^9$.

Output
For each test case, print two lines, each containing two integers: the resulting 2×2 matrix A^X. Each element should be printed modulo $10^9 + 7$.

E. Fast Series Drift

2.5 seconds, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given three integers ***a***, ***n*** and ***m***. Calculate $(a^1 + a^2 + \ldots + a^n) \% m$.

Input
The first line contains an integer ***T*** $(1 \le T \le 10^5)$ — total numbers of test cases.

In each of the next T test cases, there are three integers ***a*** $(1 \le a \le 10^6)$, ***n*** $(1 \le n \le 10^{12})$ and $(1 \le m \le 10^9)$

Output
Print one integer — the result of $(a^1 + a^2 + \ldots + a^n) \% m$.

input
3 2 5 1000 2 9 1000 1 100 30
output
62 22 10

F. Ordering Binary Tree

1 second, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

you are given an array ***A*** of size ***N*** in **increasing** order. Find an order of these N integers such that, if these integers are inserted into a Binary Search Tree (BST) one by one, the height of the resulting BST is minimized.

A Binary Search Tree is a binary tree in which each node has at most two children, referred to as the left and right child. For any node, all elements in the left subtree are smaller than the node's value, and all elements in the right subtree are greater than the node's value.

The height of a Binary Search Tree is defined as the maximum depth among all the nodes in the tree.

Note: All the elements in the array A are guaranteed to be unique. In other words, $A_i \neq A_j$ if $i \neq j$.

Input
The first line contains an integer ***N*** $(1 \leq N \leq 10^5)$ — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq 10^9)$ in non-descending order separated by spaces.

Output
Output the order of the elements such that when inserted into a Binary Search Tree, the height of the tree is minimized. If there are multiple such orders then find any of them.

input
5 1 2 3 4 5
output
3 1 2 4 5

G. 220 Trees

1 second, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

There is a Binary Tree with ***N*** nodes. You are given the in–order and pre-order traversals of the tree. Your task is to determine the post-order traversal of the tree.

Input
The first line contains an integer N $(1 \leq N \leq 1000)$ — the number of nodes in the binary tree.

In the next line, there will be N integers $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq N)$ separated by spaces – representing the in-order traversal of the tree.

The following line, there will be N integers $b_1, b_2, b_3 \ldots b_n$ $(1 \leq b_i \leq N)$ separated by spaces – representing the pre-order traversal of the tree.

Output
Print N space-separated integers representing the post-order traversal of the binary tree.

input
5 4 2 5 1 3 1 2 4 5 3
output
4 5 2 3 1

H. 220 Trees Reassessed

1 second, 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

There is a Binary Tree with ***N*** nodes. You are given the in-order and post-order traversals of the tree. Your task is to determine the pre-order traversal of the tree.

Input
The first line contains an integer N $(1 \leq N \leq 1000)$ — the number of nodes in the binary tree.

In the next line, there will be N distinct integers $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq N)$ separated by spaces – representing the in-order traversal of the tree.

The following line, there will be N distinct integers $b_1, b_2, b_3 \ldots b_n$ $(1 \leq b_i \leq N)$ separated by spaces – representing the post-order traversal of the tree.

Output
Print N space-separated integers representing the pre-order traversal of the binary tree.

input
5 1 2 3 4 5 1 4 5 3 2
output
2 1 3 5 4

input
10 1 2 3 4 5 6 7 8 9 10 2 1 3 5 6 7 4 10 9 8
output
8 4 3 1 2 7 6 5 9 10

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